DESCRIPTION OF COURSEWORK

Course Code	EEE406
Course Name	Power Systems Analysis
Lecturer	Dr Abdul Rashid Abdul Rahman
Academic Session	09/2025
Assessment Title	Laboratory 1 - Power flow studies using Gauss-Seidel Method

A. Introduction/ Situation/ Background Information

Objectives

To perform power flow studies using Gauss-Seidel method for the given tasks.

Pre-Laboratory

Knowledge on power flow and familiarity with PowerWorld Simulator software is very important prior to conducting this laboratory. Therefore, students should complete the Pre-Laboratory steps below:

1. Study the power flow analysis through lecture notes or reference book.
2. Install the PowerWorld Simulator software.
3. Learn on how to use PowerWorld Simulator software from "PowerWorld Simulator Quick Guide Ver 5".

Watch related tutorial video at powerworld.com or youtube.com websites for better understanding on using PowerWorld Simulator software.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Analyze transmission system and power flow with Gauss-Seidel and Newton-Raphson methods.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group of **four (4) or five (5)** students's laboratory to be completed in 2 weeks.

Simulation files (PowerWorld's Simulator pwb and pwd files) should be submitted with the report using filename "Lab1Report" in softcopy (PDF) for every task that should follow the following format:

1. Introduction
2. Methodology
3. Results and Discussion

Due Date: **26th December 2025 (Week 13).**

Software Tool: PowerWorld Simulator software.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Draw the two-busbar per unit representation using appropriate base values. Include all components: generator, transformers, transmission line, and convert the load to per unit. Clearly identify Bus 1 (generator) and Bus 2 (load).	20
2.	Perform manual calculation for THREE (3) iterations using Gauss-Seidel method to solve bus voltages. Show step-by-step workings, assumptions, and evaluate your results.	30
3.	Develop the two-busbar per unit system in PowerWorld. Simulate using Gauss-Seidel method for THREE (3) iterations. Capture the one-line diagram and load flow results. Compare and discuss with manual results for every iteration.	30
4.	Comment on findings from manual and simulation results. Discuss the power transferred on a single transmission line and identify key parameters that affect the ability to transfer real and reactive power over long distances.	20
	TOTAL	**100**

F. Task(s)

1. For the system shown, draw the two-busbar per unit representation by choosing appropriate base values. Consider Bus 1 to be at the Generator and Bus 2 to be at the Load point. Consider the resistances and reactances of all components, such as Generator, Transformers, Transmission Lines to be present between these two buses. Load should be converted to per unit terms and should be connected beyond Bus number 2.
2. Carry out calculations showing three iterations for finding the voltages at the two busbars. Consider appropriate assumptions and show your calculations step-by-step.
3. Verify the voltages in PowerWorld by simulating the two-busbar per unit system and capture the outputs (one-line diagram and load flow results). What are the voltage values in actual quantities (kV)?
4. Comment on your findings and comment on the power transferred on a single transmission line, i.e., what are the parameters that affect the ability of a transmission line to transfer bulk real and reactive powers over long distances?
5. Display your simulation steps and results in the report along with your discussion.

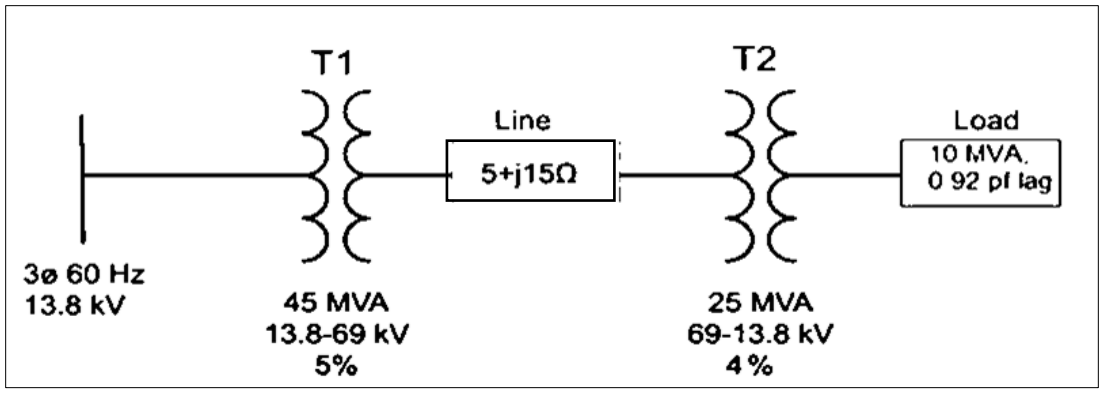


Figure 1. One line diagram of power system

APPENDIX 1

MARKING RUBRICS

Component Title	Power flow studies using Gauss-Seidel and Newton-Raphson				Percentage (%)		
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
System design	100% reliable and unique isolated network topology design and using logical parameters of industry data	80% reliable and unique isolated network topology design and using logical parameters of industry data	60% reliable and unique isolated network topology design and using logical parameters of industry data	40% reliable and unique isolated network topology design and using logical parameters of industry data	plagiarized isolated network topology design	2	
Manual calculation to solve the voltages using Gauss-Seidel method	100% detail steps of calculation shown with comprehensively and correctly evaluated	80% detail steps of calculation shown with comprehensively and correctly evaluated	60% detail steps of calculation shown with comprehensively and correctly evaluated	40% detail steps of calculation shown with comprehensively and correctly evaluated	less than 40% detail steps of calculation shown with comprehensively and correctly evaluated	3	
PowerWorld Simulator simulation using Gauss-Seidel method	100% detail simulation conducted and compared with manual calculation for every iteration	80% detail simulation conducted and compared with manual calculation for every iteration	60% detail simulation conducted and compared with manual calculation for every iteration	40% detail simulation conducted and compared with manual calculation for every iteration	less than 40% detail simulation conducted and compared with manual calculation for every iteration	3	
Overall power flow results discussion	100% detail overall power flow results being discussed.	80% detail overall power flow results being discussed.	60% detail overall power flow results being discussed.	40% detail overall power flow results being discussed.	less than 40% detail overall power flow results being discussed.	2	
					TOTAL		

Note to students: Please include the marking rubric when submitting your coursework.